

AirForce Management System

Project Overview

The AirForce Management System is a database-driven solution designed to manage airline operations during normal and crisis situations. The system helps airline staff manage flights, aircraft, pilots, passengers, baggage, disruptions, and operational alerts efficiently.

Features

1. Flight Management

Description:

Allows airline staff to create, update, and monitor flight schedules. Tracks flight status such as Scheduled, Delayed, Cancelled, or Completed.

2. Aircraft Assignment & Replacement

Description:

Stores aircraft details and assigns aircraft to flights. In emergency situations, the system supports aircraft replacement and records the changes.

3. Pilot Management

Description:

Maintains pilot information including license details, availability status, and contact information. Helps identify available pilots during operational disruptions.

4. Flight Crew Assignment

Description:

Assigns pilots to flights as Captain or Co-Pilot. Supports many-to-many relationships between flights and pilots through assignment records.

5. Passenger Booking Management

Description:

Stores passenger reservations and booking information. Tracks booking status such as Confirmed or Cancelled.

6. Passenger Rebooking System**Description:**

Automatically records passenger rebooking during flight delays or cancellations and links the passenger to a new flight.

7. Crisis & Disruption Tracking**Description:**

Records operational issues such as flight delays, pilot illness, aircraft failures, and weather disruptions. Tracks issue status until resolution.

8. Disruption Resolution Management**Description:**

Stores actions taken by airline staff to resolve operational disruptions and maintains a history of resolution activities.

9. Baggage Tracking & Recovery**Description:**

Tracks baggage movement from check-in to arrival. Identifies misdirected baggage and monitors recovery processes.

10. Alert & Notification System**Description:**

Generates operational alerts for airline staff during emergencies. Ensures timely communication and quick response to incidents.

Technologies Used

- PostgreSQL
 - SQL
 - pgAdmin 4
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Database Concepts Used

- Primary Keys
 - Foreign Keys
 - One-to-Many Relationships
 - Many-to-Many Relationships
 - Data Integrity Constraints
 - CRUD Operations
 - SQL Joins
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